Low-cost perfusion pump

Low-cost perfusion pump
Multidisciplinary design (GM1)

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Introductions!

Motivation

Collagen-based artificial vessels have applications in two key surgical contexts.

Haemodialysis

- Blood is removed, processed in a dialyser, then replaced
- Requires reliable vascular access at high flow rates
- Natural fistulas are difficult to achieve in elderly patients

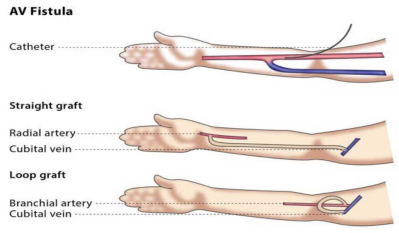
Bypass surgery

- New routes are created for blood flow
- Artificial grafts offer standardisation and greater availability

Tahmid

Discuss why haemodialysis first over bypass surgery (non life-threatening medical device, easily classification and entry into the market, and shorter duration in the body)

Arteriovenous fistula



Tahmid

Describe natural vascular changes with aging

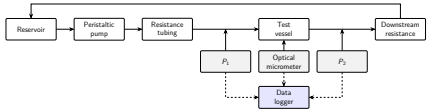
Describe iatrogenic vascular changes (e.g. repeated use of superficial veins)

Prolonged time to maturation in the elderly (normally 6–12 weeks)

Low-cost perfusion pump

Existing laboratory bench

Existing laboratory bench



Limitations

- Excess tubing creates a cluttered workspace
- Achieving 300 mL min^{-1} requires more than 250 rpm $\gg$ resting cardiac frequency range, 60–120 bpm

Luca

Went to see the current 4th year project setup

- Consists of a peristaltic pump where flow rate can be modified
- Compliance tubing to smooth out pulses from the pump
- Enters test section where the vessel diameter is tracked using an optical micrometer, and the pressure profile is detected using a differential pressure sensor
- Downstream resistance tubing provides a back pressure that changes the mean pressure of the profile

Main problem areas:

- Large offset from atmospheric pressure means pressure resolution isn't ideal
- Tubing creates a cluttered test area
- Cannot model flow rates of 300mL/min as creates non-realistic heart rates of 350bpm

Design brief

Requirements

- Continuous flow for 90 min
- Mean perfusion pressure 80–120 mmHg
- Compatible with 6 mm OD tubing
- Closed-loop control via pressure sensing
- Non-destructive to RBCs and endothelial layer

Desiderata

- Physiologically realistic pulsatile pressure profile
- Multiple pumps in parallel
- Selectable pre-programmed flow profiles
- Burst pressure ramp protocol
- Peak flow rate 300 mL min^{-1}

Luca

After the meeting with Dr Markaki, developed design brief for our pump

Split into a list of desirable and essential features

Key performance indicators are:

90 min runtime
Mean pressure between 80–120mmHg
Pressure adjustment via closed loop control
Non-destructive to RBC's

The desirable functions are:

Optimise the pressure profile so that it is physiologically realistic
Design a system so multiple pumps can be used in parallel
Have pre-programmed profiles for different vessels and ages/conditions
Test the burst pressure of the grafts by using a pressure ramp
Achieve a peak flow rate of 300ml/min

Low-cost perfusion pump

Initial design: syringe pump

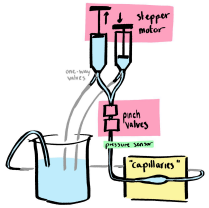
Initial design: syringe pump

2 syringes for continuous flow

Valves prevent backflow

Pinch valves/motor variation to create non-standard waveform

Pressure sensors for a feedback loop



Lucy

Project resources

item	qty.	unit cost (£)
microcontroller	1	20
power board	1	20
lead screw & stepper motor	2	15
pressure sensors	2	10
one-way valves	4	2
3/2 valves	2	10
pinch valves	2	20
syringes	2+	1
tubing	—	10
simple alphanumeric display	1	10
T-type connectors	2	1
Departmental		
3D printer	—	—
housing material	—	—
total		**∼150**

Lonnie

Microcontroller — controls the pressure profile and monitors flow rate throughout the system

Power board — converts 240V mains supply down to 5V for use by the electronics

Lead screw & stepper motor — pushes and retracts the syringe accurately to displace precise fluid volumes

Pressure sensors — measure the pressure profile within the test section to close the feedback loop

One-way valves — prevent backflow from the syringe into the test section during retraction

3/2 valves — provide an alternative to one-way valves with more flexible flow path switching

Pinch valves — create a controlled pressure drop to set the downstream resistance

Syringes — provide a consistent and controllable flow rate into the circuit

Alphanumeric display — shows real-time flow rate and minimum, peak, and average pressure readings to the user

T-type connectors — allow fluid to branch into the pressure sensors without disrupting the main flow path

Low-cost perfusion pump

Final product cost

Final product cost

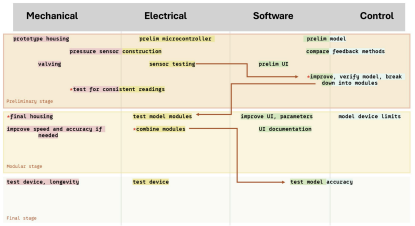
item	qty.	unit cost (£)
microcontroller	1	5–20
power board	1	10–20
lead screw & stepper motor	2	15
pressure sensors	2	3–10
valves	2–6	2
syringes	2	1
tubing	—	—
total		**50–100**
commerical syringe pumps		250–3000

Lonnie

Low-cost perfusion pump

Timeline

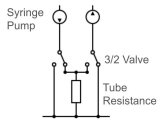
Timeline



Lonnie

Mechanical

- Design housing for motors, syringes, and reservoir
- Design and assemble connections for pressure sensors
- Design and assemble valve junctions valve junctions
- Data collection for control system parameter tuning
- Ensure final product is easy to 3D print and assemble, and that multiple pumps can be made and run at once



Lucy

Mechanical: pressure sensing function

- Pressure is the primary output variable, the physiological signal being replicated
- Closed-loop control drives the pump to track a target pressure waveform
- Mean perfusion pressure must be maintained in a range, e.g. 80–120 mmHg at rest
- The heart beats at ∼1 Hz, but fast transients in the waveform must be faithfully tracked

Lucy

Mechanical: pressure sensing specifications

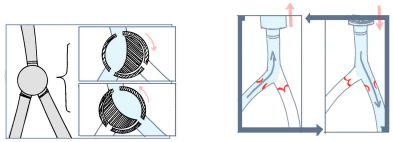
parameter	value	rationale
Operating range	80–120 mmHg	Physiological perfusion target at rest
Full range	150–500 mmHg	Burst testing and safety margins
Sensor type	Gauge	Eliminates 760 mmHg atmospheric offset to avoid sacrificing sensor dynamic range
Minimum bandwidth	100 Hz	Nyquist for estimated 50 Hz control loop sampling rate

Luca

Mechanical: valves

2 one-way valves at syringe exit with the same cracking pressure mechanism as in the heart

Optional 3/2 valve setup for more control



Lucy

Mechanical: motors

Stepper motor

- Precise position control
- Fast, inexpensive speed changes
- 200 steps/rev, 0.1 N m hold torque

Key calculations

$$n = \frac{60Q}{AL} \approx 37.5\,\mathrm{rpm} \qquad T = \frac{FL}{2\pi\eta} \approx 6.8\,\mathrm{mN\,m}$$

N.B. n: motor speed; Q: flow rate; A: syringe area; $L = 8\,\mathrm{mm}$: lead screw pitch; T: torque; F: axial force; η: efficiency

Luca

Mechanical: waveform

Simpler

- Stepper motor modulates syringe push rate directly
- Only 1 pressure sensor required

More control

- Pinch valves added downstream of syringes
- AC pressure component superimposed on DC syringe output
- 2 pressure sensors required

Luca

Electrical

Preliminary

Familiarise with microcontroller
Connect and test sensors
Compare sensor data with control team

Modular

Convert model to microcontroller
Combine modules into larger subsystems
Support mechanical team with soldering/other electronic tasks

Final

Fix errors using model with control team
Extensive testing
Select cheapest viable microcontroller given features used

Lonnie

Software

- Well-documented GUI and CLI (scripting)
- Real-time visualisations and plots
- Cross-platform
- Rich data logging with convenient export formats

Tahmid

Software: profile configuration

- Customise the target pressure profile via:
  - Physiological parameters (heart rate, systolic/diastolic pressures)
  - Demographics (age, sex)
  - Activity level (rest vs. exercise)
  - Vessel type (artery, capillary, vein)
- Set operation schedules
  - e.g. ramp pressure to test vessel integrity over time

Tahmid/India

Software: monitoring & exporting

- Live pressure trace at the test start point
  - Cumulative error (especially during syringe transition)
  - System uptime
- Automatic flagging of extreme pressure values
  - Sudden drop → vessel rupture
  - Sudden rise → blockage or clotting
- Export data traces as CSV

Tahmid

Software: modelling the pressure profile

Windkessel model to find the mean pressure for a specific vessel

The Womersley number can be used characterise the vessel (inertia dominated, as in large arteries, or viscosity dominated as in capillaries)

Carry out Fourier analysis on parameter-sensitive cardiac waveforms found in literature [**malindzakFourierAnalysisCardiovascular1970**]

$$P(t) = P_{\text{mean}} + \sum_{n=1}^{N} A_n \sin(n\omega t + \phi_n)$$

Set ω from heart rate, P_{mean} from the Windkessel model and fit $\{A_n\}$ and $\{\phi_n\}$ from literature

Tahmid

Control: objectives

Goal
Track and approach a target pressure profile in real time.

Constraints

- Must not exceed maximum safe pressure — vessel bursting is irreversible
- Must account for system non-linearities: syringe mechanics, valve dynamics

India

Not just steady state pressure, track continuously varying waveform in real time

Burst constraint is hard, overshooting once ruptures vessel — be conservative

Non-linearities — may get dead zones from switching, tubing compliance stores energy, fluid inertia lags (does not fit linear systems)

Strategy — sample rapidly so each step stays near linear

Control: solution

Linear systems negative feedback loop

- Filter raw pressure sensor output
- PID controller maps pressure error → motor speed
- High sampling frequency (∼50 Hz) minimises non-linearity effects by keeping each control step within a near-linear operating region
- Control both the DC mean pressure component and the AC pulsatile waveform

Key challenge: modelling the plant

Specifically the relationship between motor speed and output pressure.

India

Negative feedback loop — filter → PID → motor speed command

Filter first to remove high-frequency noise before computing error

PID terms:

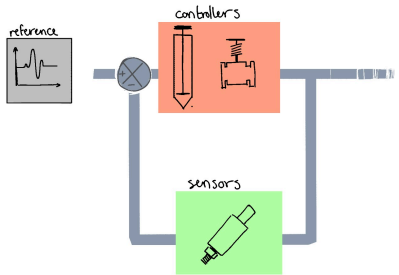
P term: immediate response to current error
I term: corrects persistent mean pressure offset
D term: damping to prevent overshoot

Controlling DC and AC simultaneously — mean pressure set by average motor speed, pulsatile shape by modulating speed cyclically using pinch valves downstream

DC and AC interact — changing mean flow rate affects valve operating point

Biggest unknown: plant model — all of the unknowns sit between motor and sensor — cannot set PID gains until this has been well characterised

Low-cost perfusion pump



India

Reference waveform comes in from software — generated as Tahmid explained

Compute error between reference and measured

Controllers block are the actual actuators (plant)

Sensors — read pressure and close the loop

Everything between motor and measurement point is in loop — why plant model is really important

Control: extensions

Compliance modelling
Burst pressure modelling
Combine deformation gradients with pressure data to predict failure point
Enables safe, more controlled burst pressure testing

India

Compliance modelling — log pressure and vessel deformation continuously to build compliance curve across operating range

Burst pressure modelling — ramp to failure can be hard to control

Use deformation gradients as indicator — vessel shows characteristic change in response as they approach failure

Track inflection point in real time — safer and reproducible burst pressure measurement

Next steps: initial prototype + testing

Mechanical

- Build syringe → valves → reservoir → syringe fluid circuit
- Build motor/syringe housing
- Connect pressure sensors to tubing

Electrical

- Learn microcontroller software

Control

- Build and test control system on simulated pressure waveforms
- Characterise controller lag and amplitude response in real time

Software

- Choose a tech stack
- Begin basic modelling features and plan UIs

Mechanical — Lucy & Luca

Control — India

Run PID against simulated waveforms to develop gains before even touching vessel
Open loop tests once mechanical circuit is ready to characterise plant model

Electrical — Lonnie

Software — Tahmid

Low-cost perfusion pump

Fallback: if syringe system doesn't work as planned

Gravity-powered flow

Diaphragm for pressure pulses

Lonnie